a **Three builds of the same screens, one pool of records**

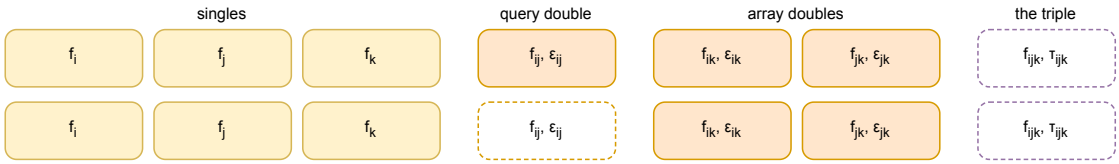
010 build: triples only
the 376,732 Kuzmin 2018 and 2020 trigenic records, one value per label (the mean over sources); no singles, no doubles. The model sees only gene identity and the triple's own score.

025 closure pool S3, labels merged
every single (5,694), every double whose gene pair lies inside some triple (739,236) and every triple (376,732): 1,121,645 records. One value per label, the mean over sources, so a converted 0 can average with a measurement.

030 per-entry pool, nothing merged
the same pool, 1,121,662 records, keeps every source entry: 4,860,168 entry rows (23,509 on singles, 4,054,063 on doubles, 782,596 on triples), each tagged with its source dataset, and the double-mutant query strain's own fitness beside the array screen's.

b **One triple {i, j, k} of query pair ij and array gene k, the records the published identity reads, and which of them train**

$\tau_{ijk} = f_{ijk} - f_{ij}\, f_k - \varepsilon_{ik} - \varepsilon_{jk}$ filled: a training record white, dashed: withheld from training



On both arms the held-out triple's own fitness and score leave together, since a record carries all of a genotype's entries. On arm Q the held-out query pair's double leaves as well; the array gene, the singles and the array-side doubles stay.

c **The two arms, every count from the split artifacts**

Arm R: random over triples
the 010 split carried by gene-set identity: 301,386 train, 37,673 validation, 37,673 test triples; every single and every closure double trains; a held-out triple's query pair, array gene and all six lower-order records are in training.

Arm Q: query-pair disjoint
420 recurring query pairs, 89 held out with every triple they appear in: 301,236 train, 37,705 validation, 37,791 test triples; the 88 doubles whose pair is held out leave training, so f_{ij} and ε_{ij} are never seen; array genes are shared across the split.